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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/298,208	05/28/2021	Hiroshi ITO	110882.PE657US	9568

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EXAMINER	
CASTRO, PAUL A	

ART UNIT	PAPER NUMBER
3662	

NOTIFICATION DATE	DELIVERY MODE
09/29/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/298,208	Applicant(s) ITO et al.	
	Examiner PAUL A CASTRO	Art Unit 3662	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on RCE filed 09/10/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 13-23 is/are pending in the application.

5a) Of the above claim(s) 16-23 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 13-15 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☑ Claim(s) 16-23 are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 05/28/2021 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

This action is in response to the RCE filed on 09/10/2024.

Claims 13-23 are pending.

Claims 16-23 are withdrawn (with traverse)

Continued Examination Under 37 CFR 1.114

A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 09/10/2024 has been entered.

Response to Arguments

1. Applicant's arguments with respect to claims have been considered but are moot because the arguments do not apply to any of the references being used in the current rejection.

Election/Restrictions

Claims 16-23 withdrawn from further consideration pursuant to 37 CFR 1.142(b), as being drawn to a nonelected species B, there being no allowable generic or linking claim. Applicant timely traversed the restriction (election) requirement in the reply filed on 09/10/2024.

Election was made by original presentation, with traverse of Species A in the reply filed on 09/10/2024 is acknowledged. The traversal is on the ground(s) that there is not a search

burden due to the independent claims' last clause being the only difference. This is not found persuasive because although the first few clauses of the differing independent claims are very similar, they are also broadly claimed. When the claims get to the last clause, this is where the crux appears most prevalent. As this appears to be the case, the differing last clause is the more important components when searching for prior art. There are thousands of different arts in vehicle systems alone, and defining what kind of data vs quality of data as criteria for decision making would generate different searches. If Examiner found a system that made a system determination based on shapes of vehicles or speed, this would not read on claims 13-15's quality of signals for determination and vice versa. This would generate a search burden on Examiner as it would lead to additional different search criteria between the differing claims. This is further evident by Applicant's use of drawings which discloses the different control schemes, exemplified in Fig. 6-7 and Fig. 11.

The requirement is still deemed proper and is therefore made FINAL.

Claim Objections

The objection is withdrawn.

112 Rejection

This rejection is withdrawn.

103 Rejection

The Applicant remarks for claim 13 that the prior art does not read on the current amended claim. The Examiner disagrees by the following rationale.

More specifically the Applicant cites the limitations of, "radio wave signal being stronger than or equal to a threshold value and being greater than a radio wave signal received from the other of the first and the second". Applicant refers to the citations used for previous prosecution. First with prior art Horn, it has been previously established that the "center vehicle" of Horn for all intents and purposes, functions the same as Applicants "following vehicle" [see previous round of prosecution for more details] with dependence of bilateral communication with other vehicles. Next Horn does state [¶ 109] with augmenting different vehicle sensor data with each other:

One advantage of this approach is robustness due to redundancy, since the failure of a wireless link does not disable the bilateral control system of the center vehicle. If a combination of sensors on the moving vehicles and communication from leading and following vehicles is used, then a failure of an individual information source does not perturb the stability of the overall traffic flow and the suppression of traffic flow instabilities at all.

First, the stating "failure of a wireless link" would indicate a minimum threshold required to used said communication link (a radio wave signal stronger/greater than a threshold, to be used in communication). Next, Horn states that having a failure of communication, does not disable bilateral control system of the center vehicle, instead vehicle data is used from other following/leading vehicles, as in the communication used is further based on which vehicles have the better ability to communicate at the time. Horn then states, "A failure of individual information source does not perturb the stability." As in Horn uses the best available communication with other vehicles in the formation, when at least compared to failure to communicate with a first vehicle (then get information from an alternate vehicle that has better communication available.

Prior art Horn is understood to disclose a system of platooning vehicles as claimed in the limitations.

Claim 14 discloses similar limitations and the claims remain rejected by similar rationale above.

Claim 15 does not disclose the same language as remarked by the Applicant nor does it disclose any similar amendments. As such it is understood that claim 15 rejection is not remarked on with relation to the prior art.

Claim 15 rejections are maintained.

Examiner's Note

Examiner has cited particular paragraphs / columns and line numbers or figures in the references as applied to the claims below for the convenience of the applicant. Although the specified citations are representative of the teachings in the art and are applied to the specific limitations within the individual claim, other passages and figures may apply as well. It is respectfully requested from the applicant, in preparing the responses, to fully consider the references in entirety as potentially teaching all or part of the claimed invention, as well as the context of the passage as taught by the prior art or disclosed by the examiner. Applicant is reminded that the Examiner is entitled to give the broadest reasonable interpretation to the language of the claims. Furthermore, the Examiner is not limited to Applicants' definition which is not specifically set forth in the claims.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 13-14 are rejected under 35 U.S.C. 103 as being unpatentable over US 20140236449 ("Horn").

As per claim 13 Horn a vehicle control apparatus mounted on a third vehicle in a vehicle following running system in which a first vehicle, a second vehicle subsequent to the first vehicle, and the third vehicle subsequent to the second vehicle carry out following running by being non- mechanically connected **[¶ 27 vehicle controller system provides an output to the mechanism to control forward motion…. As a function of the outputs of the one or more leading distance sensors, of the one or more leading speed sensors, ¶ 78 Of course, the leading and following vehicles can have such control systems, Fig. 1 (see also Response to arguments for 103 rejection above)**],

the vehicle control apparatus being configured to:

selectively acquire first vehicle information including a first physical amount regarding a motion of the first vehicle and second vehicle information including a second

physical amount regarding a motion of the second vehicle **[¶ 22 "adaptive cruise control" rely on information about a leading vehicle, ¶ 23 front to back controls… based on speed and distance (known in the art), ¶ 24 a center vehicle controls both front and following vehicles… all vehicles are provided with the control system ¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 1 vehicles F, C, and L, Fig. 4 (input) of leading distance and speed sensors (understood to come from more than one leading vehicle as explained above)**],

determine a third physical amount regarding a motion of the third vehicle based on the vehicle information acquired selectively from the first vehicle information and the second vehicle information**[¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 4 forward motion control of vehicle]**, and

output an instruction based on the determined third physical amount to a braking apparatus or driving apparatus and a steering apparatus of the third vehicle **[¶ 27 can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands]**, wherein

based on the instruction, the driving apparatus applies a driving torque, the braking apparatus applies a hydraulic pressure, and a steering apparatus applies a steering angle, thereby controlling the braking or driving and steering of the third vehicle [¶ **28 provide fuel and braking commands**], and

in the selective acquisition of vehicle information, the third vehicle selectively receives vehicle information from one of the first vehicle and the second vehicle from which a radio wave signal is received by the third vehicle in wireless communication of the vehicle information, the radio wave signal being stronger than or equal to a threshold value and being stronger than radio wave signals received from the other of the first and the second vehicle **[¶ 27 can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 109 Augments the information from sensors on the center vehicle with information from sensors in the leading and following vehicles… advantage of this approach is robustness due to redundancy, since the failure of a wireless link (predetermined condition, any failure of communication link, can use other communication links, such as those in the other vehicle instead)],**

the received radio wave signal is associated with the vehicle information sent by the wireless communication **[¶ 107 … information may, for example, be transmitted from the leading and following vehicles using radio frequency (RF) communication…]**.

Horn describes the vehicle control system as a leading vehicle, center control vehicle, and a following vehicle. Horn further describes the control vehicle as handling sensor data of both the front leading vehicle and back following vehicle. Horn then states that the "control vehicle" label is for purposes of explaining the invention and avoid ambiguity and however, that all three

vehicles possess the vehicle control system and are all capable of bi-direction communication for the control system **[¶ 78]**.

Horn describes a different order of vehicles however; it would have been obvious to one of ordinary skill in the art before the effective filing date the invention was made to understand the vehicle control system of Horn would include accepting data from vehicles in any order for purposes of control automation of vehicle processions to minimize different acceleration jerkiness between vehicles to increase vehicle comfort and efficiency.

As per claim 14 Horn discloses a vehicle control apparatus mounted on a third vehicle in a vehicle following running system in which a first vehicle, a second vehicle subsequent to the first vehicle, and the third vehicle subsequent to the second vehicle carry out following running by being nonmechanically connected **[¶ 27 vehicle controller system provides an output to the mechanism to control forward motion…. As a function of the outputs of the one or more leading distance sensors, of the one or more leading speed sensors, ¶ 78 Of course, the leading and following vehicles can have such control systems, Fig. 1 (see also Response to arguments for 103 rejection above)]**,

the vehicle control apparatus being configured to:

selectively acquire first vehicle information including a first physical amount regarding a motion of the first vehicle and second vehicle information including a second physical amount regarding a motion of the second vehicle**[¶ 22 "adaptive cruise control" rely on information about a leading vehicle, ¶ 23 front to back controls… based on speed and distance (known in the art), ¶ 24**

a center vehicle controls both front and following vehicles… all vehicles are provided with the control system ¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 1 vehicles F, C, and L, Fig. 4 (input) of leading distance and speed sensors (understood to come from more than one leading vehicle as explained above)],

determine a third physical amount regarding a motion of the third vehicle based on the vehicle information acquired selectively from the first vehicle information and the second vehicle information**[¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 4 forward motion control of vehicle**],, and

output an instruction based on the determined third physical amount to a braking apparatus or driving apparatus and a steering apparatus of the third vehicle **[¶ 27 can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands]**, wherein

based on the instruction, the driving apparatus applies a driving torque, the braking apparatus applies a hydraulic

pressure, and a steering apparatus applies a steering angle, thereby controlling the braking or driving and steering of the third vehicle [¶ **28 provide fuel and braking commands**],

in the selective acquisition of vehicle information, the third vehicle selectively receives vehicle information from one of the first vehicle and the second vehicle from which a radio wave signal is received by the third vehicle in wireless communication, the radio wave signal being stronger than or equal to a threshold value and coming from said one vehicle ahead of the other **[¶ 27 can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 109 Augments the information from sensors on the center vehicle with information from sensors in the leading and following vehicles… advantage of this approach is robustness due to redundancy, since the failure of a wireless link (predetermined condition, any failure of communication link, can use other communication links, such as those in the other vehicle instead)]** and

the received radio wave signal is associated with the vehicle information sent by the wireless communication **[¶ 107 … information may, for example, be transmitted from the leading and following vehicles using radio frequency (RF) communication…]**.

Horn describes the vehicle control system as a leading vehicle, center control vehicle, and a following vehicle. Horn further describes the control vehicle as handling sensor data of both the front leading vehicle and back following vehicle. Horn then states that the "control vehicle" label is for purposes of explaining the invention and avoid ambiguity and however, that all three

vehicles possess the vehicle control system and are all capable of bi-direction communication for the control system **[¶ 78]**.

Horn describes different order of vehicles however; it would have been obvious to one of ordinary skill in the art before the effective filing date the invention was made to understand the vehicle control system of Horn would include accepting data from vehicles in any order for purposes of control automation of vehicle processions to minimize different acceleration jerkiness between vehicles to increase vehicle comfort and efficiency.

Claims 15 are rejected under 35 U.S.C. 103 as being unpatentable over US 20140236449 ("Horn") in further view of US 20120147802 ("Ukita").

As per claim 15 Horn discloses a vehicle control apparatus mounted on a third vehicle in a vehicle following running system in which a first vehicle, a second vehicle subsequent to the first vehicle, and the third vehicle subsequent to the second vehicle carry out following running by being nonmechanically connected **[¶ 27 vehicle controller system provides an output to the mechanism to control forward motion…. As a function of the outputs of the one or more leading distance sensors, of the one or more leading speed sensors, ¶ 78 Of course, the leading and following vehicles can have such control systems, Fig. 1 (see also Response to arguments for 103 rejection above)],**

the vehicle control apparatus being configured to:

selectively acquire first vehicle information including a first physical amount regarding a motion of the first vehicle and second vehicle information including a second physical amount regarding a motion of the second vehicle**[¶ 22 "adaptive cruise control" rely on information**

about a leading vehicle, ¶ 23 front to back controls… based on speed and distance (known in the art), ¶ 24 a center vehicle controls both front and following vehicles… all vehicles are provided with the control system ¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 1 vehicles F, C, and L, Fig. 4 (input) of leading distance and speed sensors (understood to come from more than one leading vehicle as explained above)],

determine a third physical amount regarding a motion of the third vehicle based on the vehicle information acquired selectively from the first vehicle information and the second vehicle information **[¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 4 forward motion control of vehicle**], and

output an instruction based on the determined third physical amount to a braking apparatus or driving apparatus and a steering apparatus of the third vehicle **[¶ 27 can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands**], wherein

based on the instruction, the driving apparatus applies a driving torque, the braking apparatus applies a hydraulic pressure, and a steering apparatus applies a steering

angle, thereby controlling the braking or driving and steering of the third vehicle [**¶ 28 provide fuel and braking commands**], and

in the selective acquisition of vehicle information, the third vehicle selectively receives vehicle information from one of the first vehicle and the second vehicle, wherein the vehicle information has a time delay in wireless communication that is shorter than or equal to a threshold value **[¶ 108 sensors on the center vehicle are replaced by sensors in the leading… vehicles… a system using information conveyed from leading and following vehicles may benefit from the potentially higher accuracy and timeliness of the information** (*improved timeliness is understood as time delays in wireless communications is considered and having multiple sensor data links from different vehicles are available, with the advantage being having an option to choose a more timely data source*)**, ¶ 109 One advantage is robustness due to redundancy, since the failure of a wireless link does not disable the bilateral control… if combination of sensors… from leading… vehicles is used, then failure of an individual information source does not perturb the stability** (*failure of sensors is understood as a threshold to use one communication link over another*)] and that a higher timeliness is considered in wireless communication between the other of the first and the second vehicle and the third vehicle **[¶ 108 sensors on the center vehicle are replaced by sensors in the leading… vehicles… a system using information conveyed from leading and following vehicles may benefit from the potentially higher accuracy and timeliness of the information** (*improved timeliness is understood as time delays in wireless communications is considered and having multiple sensor data links from different*

vehicles are available, advantage is having an option to choose a more timely data sources)**].**

Horn describes the vehicle control system as a leading vehicle, center control vehicle, and a following vehicle. Horn further describes the control vehicle as handling sensor data of both the front leading vehicle and back following vehicle. Horn then states that the "control vehicle" label is for purposes of explaining the invention and avoid ambiguity and however, that all three vehicles possess the vehicle control system and are all capable of bi-direction communication for the control system **[¶ 78]**.

Horn describes different order of vehicles however; it would have been obvious to one of ordinary skill in the art before the effective filing date the invention was made to understand the vehicle control system of Horn would include accepting data from vehicles in any order for purposes of control automation of vehicle processions to minimize different acceleration jerkiness between vehicles to increase vehicle comfort and efficiency.

Horn is not explicit to using the shorter time delay in wireless communication.

Ukita discloses a threshold time delay **[¶ 47 the calculated transmission period is longer than the shortest allowable delay time in the allowable delay times]** and using the shorter time delay in wireless communication **[¶ 47 the communication device determines the shortest allowable delay time as the transmission period]**.

It would have been obvious to one of ordinary skill in the art before the effective filing date the invention was made to modify Horn with the teachings of Ukita to use the most efficient wireless data pathway to increase data transmission efficiency by determining fastest transmissions for the most up to date data information transfer for improved critical data updates.

Additional Art to Consider

Application **Pub. No. 20170168503** titled *AUTONOMOUS VEHICLE TOWING SYSTEM AND METHOD,* further includes a system inter-vehicle communications used for autonomous vehicle follow systems. **This is similar to the Applicant's invention in that Applicant's** in using wireless non-mechanical connections to commutate between vehicles for a leading/following relation between said vehicles.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to PAUL A CASTRO whose telephone number is (571)272-4836. The examiner can normally be reached 10-6pm on campus.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jelani Smith can be reached on 5712703969. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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